Classification of Fetal Health from Cardiotocography Using Multinomial and Tree-Based Models

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I. INTRODUCTION

Cardiotocography (CTG) is a non-invasive, widely used monitoring technique that records fetal heart rate (FHR) and uterine contractions during pregnancy and labor. CTG traces can provide early warning of hypoxia or other complications, but manual interpretation is subjective and time-consuming and can vary across clinicians. As electronic health records and labeled CTG datasets have become more available, there is growing interest in using machine learning models to support objective classification of fetal health status. Recent work has used CTG-derived features to build predictive models that distinguish normal from non-reassuring fetal status and may help clinicians triage patients more effectively.

The goal of this project is to use CTG-derived numerical features to predict fetal health status as one of three classes: Normal, Suspect, or Pathological. Because misclassifying Suspect and especially Pathological cases may have higher clinical consequences than misclassifying Normal cases, model evaluation emphasizes not only overall accuracy but also class-specific performance for the minority outcome classes.

II. METHODS & MATERIALS

A. Data Set

The data comes from the Fetal Health Classification dataset hosted on Kaggle. The dataset consists of 2,126 cardiotocography examinations from singleton pregnancies, summarized into numeric features extracted from fetal heart rate tracings and uterine contraction signals. Each record is labeled by three expert obstetricians into one of three mutually exclusive fetal health classes: Normal, Suspect, or Pathological.

The original data are anonymized and do not include patient-level demographic information, but represent a convenience sample of CTG recordings collected in routine clinical practice.

For this project, I used the Kaggle CSV file fetal_health.csv. After reading the data into R, I standardized variable names to snake_case and recoded the outcome variable fetal_health from numeric codes (1, 2, 3) to labeled factor levels ("Normal", "Suspect", "Pathological").

The dataset includes 21 continuous predictors describing fetal heart rate baseline, accelerations, fetal movements, uterine contractions, decelerations, short-term and long-term variability, and several histogram-based features of the FHR signal (e.g., histogram width, minimum, maximum, variance, and mode). A complete missingness check confirmed that there were no missing values for any predictor or outcome.

B. Study Population

In this secondary data analysis, the effective "study population" consists of CTG examinations rather than individual pregnant people, because no subject-level identifiers are included. The final analytic dataset contained 2,126 CTG records. Of these, 1,655 (77.8%) were labeled as *Normal*, 295 (13.9%) as *Suspect*, and 176 (8.3%) as *Pathological*. This strong class imbalance reflects the fact that most pregnancies are uncomplicated, but accurately identifying the smaller groups of *Suspect* and *Pathological* cases is critical for preventing adverse outcomes.

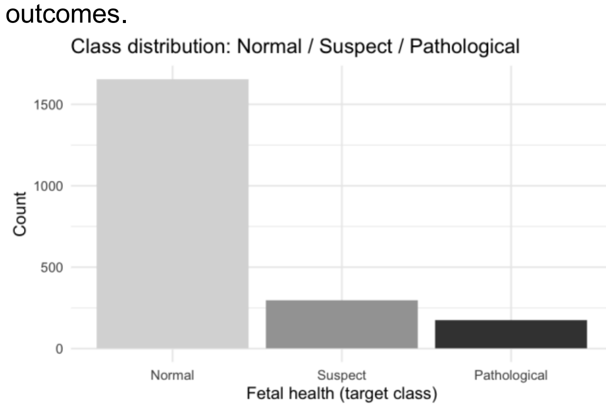

Figure 1: Distribution of fetal health classes

To support model training and evaluation, I randomly split the data into an 80% training set and 20% test set, stratified by fetal health class to preserve these proportions. The training set included 1,319 Normal, 238 Suspect, and 143 Pathological exams, while the test set included 336 Normal, 57 Suspect, and 33 Pathological exams. Exploratory boxplots illustrated that non-normal fetuses tended to have higher abnormal short-term variability, more time with abnormal long-term variability, and different uterine contraction profiles compared to Normal fetuses.

C. Statistical Methods

Table 1: Baseline Characteristics

Measurement	n	mean	sd	median	min	max
abnormal_short_term_variability	2126	46.990	17.193	49.000	12.0	87.000
accelerations	2126	0.003	0.004	0.002	0.0	0.019
baseline_value	2126	133.304	9.841	133.000	106.0	160.000
fetal_movement	2126	0.009	0.047	0.000	0.0	0.481
histogram_max	2126	164.025	17.944	162.000	122.0	238.000
histogram_mean	2126	134.611	15.594	136.000	73.0	182.000
histogram_median	2126	138.090	14.467	139.000	77.0	186.000
histogram_min	2126	93.579	29.560	93.000	50.0	159.000
histogram_mode	2126	137.452	16.381	139.000	60.0	187.000
histogram_number_of_peaks	2126	4.068	2.949	3.000	0.0	18.000
histogram_number_of_zeroes	2126	0.324	0.706	0.000	0.0	10.000
histogram_tendency	2126	0.320	0.611	0.000	-1.0	1.000
histogram_variance	2126	18.808	28.978	7.000	0.0	269.000
histogram_width	2126	70.446	38.956	67.500	3.0	180.000
light_decelerations	2126	0.002	0.003	0.000	0.0	0.015
mean_value_of_long_term_variability	2126	8.188	5.628	7.400	0.0	50.700
mean_value_of_short_term_variability	2126	1.333	0.883	1.200	0.2	7.000
percentage_of_time_with_abnormal_long_term_variability	2126	9.847	18.397	0.000	0.0	91.000
prolongued_decelerations	2126	0.000	0.001	0.000	0.0	0.005
severe_decelerations	2126	0.000	0.000	0.000	0.0	0.001
uterine_contractions	2126	0.004	0.003	0.004	0.0	0.015

All analyses were conducted in R (version 4.3.0) using the tidymodels framework for modeling and resampling. Continuous predictors were first examined descriptively using summary statistics, correlation heatmaps, and boxplots stratified by fetal health class. To prepare the data for modeling, I created two preprocessing recipes: a *scaled* recipe that removed zero-variance predictors and standardized all numeric predictors to mean 0 and standard deviation 1, and an *unscaled* recipe that only removed zero-variance predictors. The scaled recipe was used for generalized linear models, while the unscaled recipe was used for tree-based methods, which are invariant to monotone transformations.

For model development, I used the 80% training set with 5-fold cross-validation, stratified by fetal health class. Performance during tuning was evaluated using overall accuracy and balanced accuracy, defined as the average of class-specific recall across the three outcome categories.

I fit four multiclass classification models:

1. Multinomial logistic regression (nnet) using the scaled predictors.
2. Penalized multinomial logistic regression (glmnet) with an elastic-net penalty; the penalty strength and mixing parameter were tuned over a grid of values using cross-validation.
3. Random forest (ranger) using 1,000 trees and default mtry, trained on the unscaled predictors, with impurity-based variable importance.
4. Gradient boosted trees (XGBoost) with tuned hyperparameters including number of trees, tree depth, learning rate, minimum node size, and subsampling rates, using a Latin-hypercube grid and cross-validation to maximize balanced accuracy.

After selecting the best-tuned penalized and XGBoost models based on cross-validated balanced accuracy, I refit each model on the full training set and evaluated performance on the held-out 20% test set. For each model, I computed overall accuracy and the class-specific sensitivity for the Suspect and Pathological categories using the test-set confusion matrix. For the tree-based models, I also examined variable importance plots to identify which CTG features contributed most strongly to the classification.

III. RESULTS

Descriptive Findings

Across all records, the fetal heart rate baseline ranged from 106 to 160 beats per minute (bpm), with a mean of about 133 bpm. Abnormal short-term variability had a median of 49% of recording time, while abnormal long-term variability tended to be present for a smaller fraction of time but with a wide right-skewed distribution.

The correlation matrix revealed strong associations among several histogram-based features (e.g., histogram width, minimum, maximum, and variance).

Boxplots of selected predictors by class showed that *Suspect* and *Pathological* fetuses generally had higher abnormal short-term variability than *Normal* fetuses.

Model Performance

Table 2: Model Comparison

Model	Metric	Estimate
XGBoost	Accuracy	0.9624413
Random Forest	Accuracy	0.9577465
Penalized Multinomial (glmnet)	Accuracy	0.9107981
Multinomial (nnet)	Accuracy	0.9084507

On the held-out test set, all four models achieved relatively high overall accuracy, but performance varied in their ability to capture the minority classes. The multinomial logistic regression (nnet) model reached an accuracy of 0.91, with sensitivity of 0.70 for *Suspect* and 0.76 for *Pathological* case. The penalized multinomial (glmnet) model performed similarly, with an accuracy of 0.91 and slightly higher sensitivities (0.72 for *Suspect* and 0.81 for *Pathological*). Tree-based methods performed better overall. The random forest achieved an accuracy of 0.96, with very high sensitivity for both minority classes: 0.92 for *Suspect* and 1.00 for *Pathological* cases. The best XGBoost model provided the highest overall accuracy of 0.96 as well, with sensitivity 0.91 for *Suspect* and 1.00 for *Pathological*.

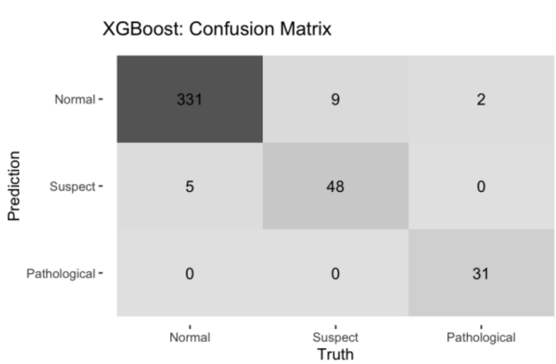

Figure 2: XGBoost confusion matrix

The XGBoost confusion matrix showed that, among 57 Suspect exams, 48 were correctly classified and 9 were misclassified as Normal; among 33 Pathological exams, 31 were correctly identified and only 2 were misclassified as Normal, with no Suspect/Pathological cross-misclassification. Both tree-based models clearly outperform the multinomial models in detecting non-normal fetuses, while maintaining excellent accuracy for Normal cases.

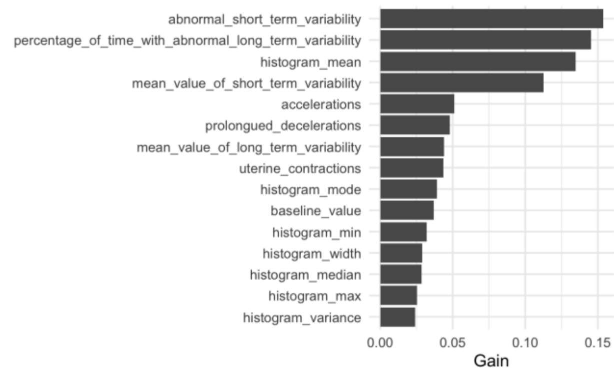

Figure 3: XGBoost Variable Importance

Variable importance plots for the chosen XGBoost models indicated that abnormal short-term variability, percentage of time with abnormal long-term variability, (FHR) histogram mean, and mean value of short-term variability were among the most influential predictors of fetal health.

IV. CONCLUSIONS

In this analysis of 2,126 cardiotocography examinations, tree-based machine learning methods were able to classify fetal health into *Normal*, *Suspect*, and *Pathological* categories with high accuracy. Both random forest and XGBoost achieved approximately 96% overall accuracy and, importantly, demonstrated excellent sensitivity for detecting Suspect and Pathological fetuses, correctly identifying nearly all Pathological cases in the held-out test set. In contrast, multinomial logistic regression models, while still reasonably accurate, showed lower sensitivity for these high-risk classes.

These findings suggest that tree-based ensemble models, especially gradient boosting, provide a promising approach for automated classification of fetal health status from CTG-derived features. If validated prospectively and integrated into clinical workflows, such models could help flag non-reassuring CTG patterns earlier and support clinicians in prioritizing closer monitoring or timely interventions to prevent adverse perinatal outcomes.

V. DISCUSSION

This project demonstrates how routinely collected CTG data can be leveraged to build machine learning models that classify fetal health status with high accuracy. I found that models incorporating nonlinear relationships and interactions among features, such as random forests and XGBoost, outperformed simpler multinomial logistic regression models.

From a methodological perspective, using stratified train/test splits and cross-validated tuning with balanced accuracy helped address the substantial class imbalance in the dataset. By focusing on test-set sensitivity for the Suspect and Pathological classes, the evaluation emphasized a clinically relevant trade-off: minimizing missed high-risk fetuses while maintaining acceptable performance for the much larger group of Normal cases. The similar performance of random forest and XGBoost suggests that both approaches are viable, though XGBoost may offer slightly better calibration or flexibility if additional tuning and feature engineering are performed.

This study has several limitations. First, the dataset is retrospective and aggregated at the level of summary CTG features, without access to detailed clinical outcomes such as neonatal intensive care unit admission or long-term morbidity. Second, the model was developed and evaluated on a single dataset; external validation on independent CTG data from other institutions would be necessary before considering clinical implementation. Third, because only CTG-derived features were available, the models do not account for maternal risk factors (e.g., age, comorbidities) or obstetric variables (e.g., gestational age), which could further improve discrimination and risk stratification. Future work could extend this analysis by incorporating raw CTG signals, additional clinical predictors, and learning approaches that explicitly weight errors in high-risk classes. Another major challenge in this dataset is the imbalanced outcome distribution, where Normal cases dominate. This makes overall accuracy alone potentially misleading; therefore, minority-class sensitivity (Suspect/Pathological) and balanced metrics are essential for evaluation. Tree-based models performed best, likely because they capture nonlinear relationships and interactions between CTG features and are less sensitive to multicollinearity than standard multinomial regression.

REFERENCES

1. Kaggle. “Fetal Health Classification” [online dataset]. Available at: https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification. Accessed December 2025.

2. World Health Organization. “Maternal mortality ratio (per 100 000 live births)”. WHO Global Health Observatory. 2024

Appendix

I. Source Data File

The source dataset is the Fetal Health Classification CTG dataset obtained from Kaggle, downloaded as fetal_health.csv and stored locally for analysis. The data include 21 numeric CTG-derived predictors and one categorical outcome variable (fetal_health) with three levels (Normal, Suspect, Pathological).

II. Data set

Fetal_health.csv

III. R Code

final-project-203a.Rmd